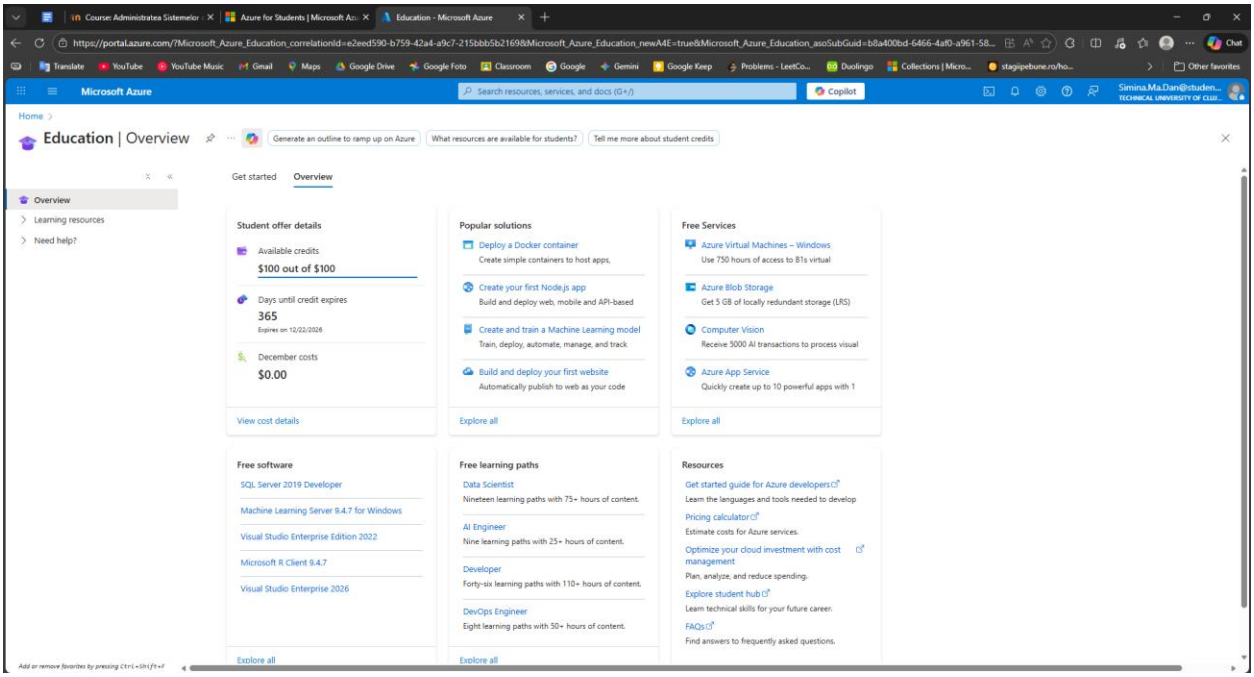
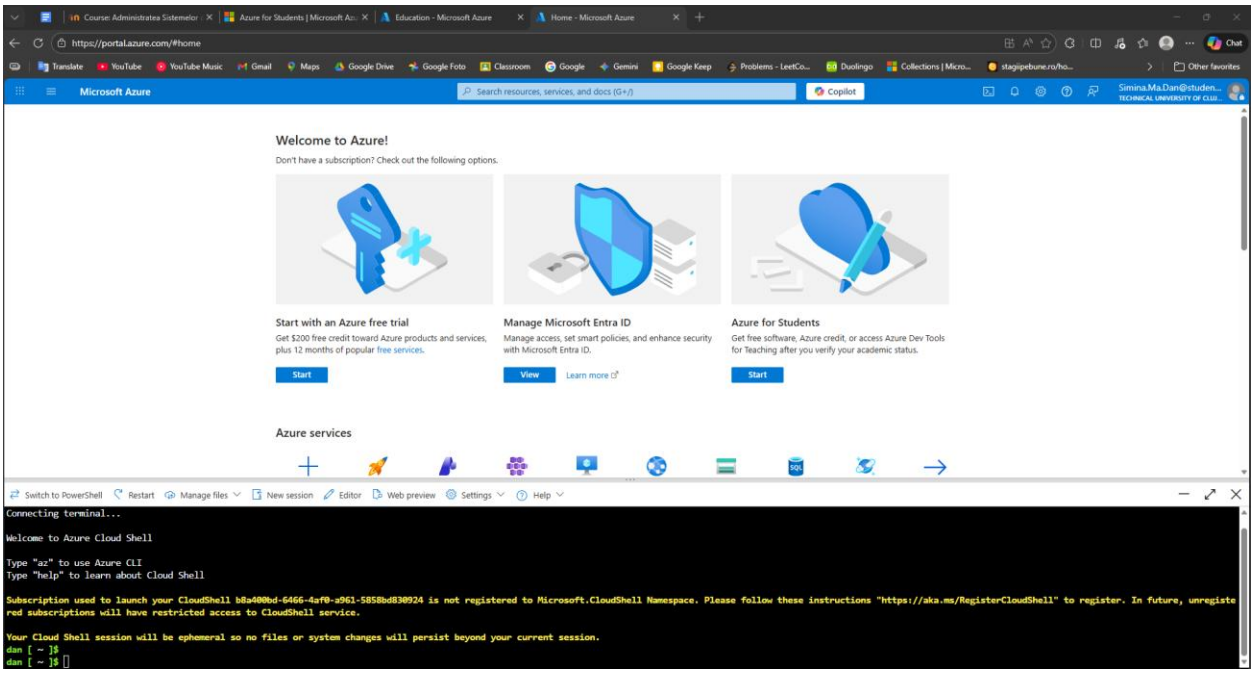
1 Preconditii

1.1 Azure pentru studenti



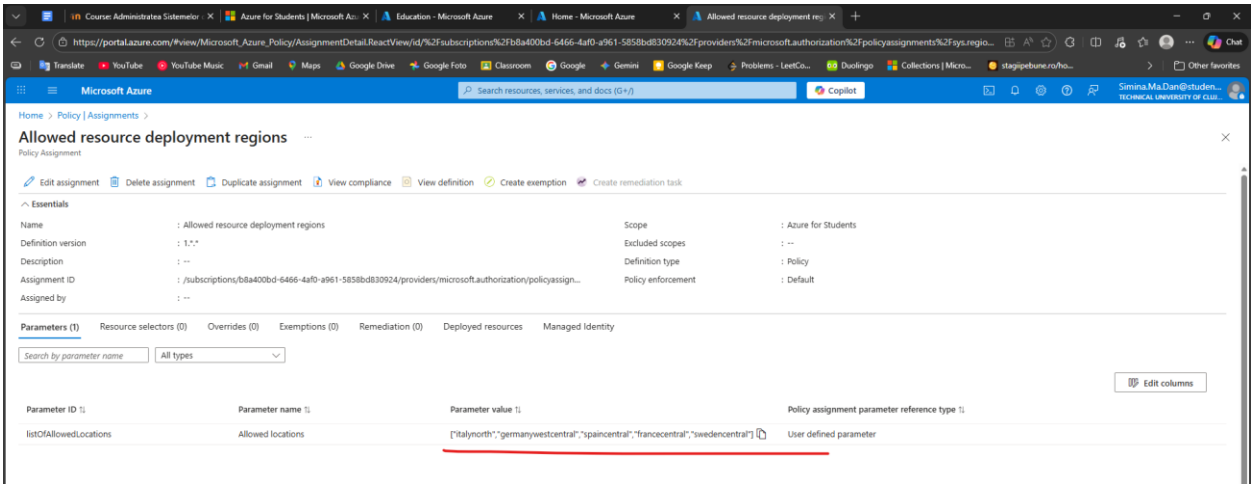
1.2 Configurarea mediului de lucru



2 Infrastructura Cloud

2.1 Zone si regiuni

2.1.1 Identificarea zonelor disponibile pentru subscriptie



2.2 Grupuri de resurse

2.2.1 Crearea unui grup de resurse

```
dan [ ~ ]$ az group create --name CloudLabRG --location italynorth
{
  "id": "/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/CloudLabRG",
  "location": "italynorth",
  "managedBy": null,
  "name": "CloudLabRG",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
dan [ ~ ]$
```

2.2.2 Stergerea unui grup de resurse

```
dan [ ~ ]$ az group delete --name CloudLabRG --yes --no-wait
dan [ ~ ]$ az group list
[]
dan [ ~ ]$
```

2.3 Masina virtuala (Virtual Machine- VM)

```
dan [ ~ ]$ az vm list-sizes --location italynorth --output table
This command has been deprecated and will be removed in a future release. Use 'az vm list-skus' instead.
MaxDataDiskCount    MemoryInMB    Name                      NumberOfCores    OsDiskSizeInMB    ResourceDiskSizeInMB
------------------  ------------  ------------------------  ---------------  ----------------  ----------------------
16                  1894400       Standard_ND96isr_H200_v5  96               1047552           1048576
8                   4096          Standard_D2pls_v6         2                1047552           0
12                  8192          Standard_D4pls_v6         4                1047552           0
24                  16384         Standard_D8pls_v6         8                1047552           0
48                  32768         Standard_D16pls_v6        16               1047552           0
64                  65536         Standard_D32pls_v6        32               1047552           0
64                  98304         Standard_D48pls_v6        48               1047552           0
64                  131072        Standard_D64pls_v6        64               1047552           0
64                  196608        Standard_D96pls_v6        96               1047552           0
8                   8192          Standard_D2pds_v6         2                1047552           0
12                  16384         Standard_D4pds_v6         4                1047552           0
24                  32768         Standard_D8pds_v6         8                1047552           0
48                  65536         Standard_D16pds_v6        16               1047552           0
64                  131072        Standard_D32pds_v6        32               1047552           0
64                  196608        Standard_D48pds_v6        48               1047552           0
64                  262144        Standard_D64pds_v6        64               1047552           0
64                  393216        Standard_D96pds_v6        96               1047552           0
8                   4096          Standard_D2plds_v6        2                1047552           0
12                  8192          Standard_D4plds_v6        4                1047552           0
```

2.3.2 Imagini ale sistemului de operare

```
dan [ ~ ]$ az vm image list
You are viewing an offline list of images, use --all to retrieve an up-to-date list
[
  {
    "architecture": "x64",
    "offer": "CentOS",
    "publisher": "OpenLogic",
    "sku": "8_5-gen2",
    "urn": "OpenLogic:CentOS:8_5-gen2:latest",
    "urnAlias": "CentOS85Gen2",
    "version": "latest"
  },
  {
    "architecture": "x64",
    "offer": "debian-11",
    "publisher": "Debian",
    "sku": "11-backports-gen2",
    "urn": "Debian:debian-11:11-backports-gen2:latest",
    "urnAlias": "Debian11",
    "version": "latest"
  }
```

2.3.3 Crearea unui VM

```
dan [ ~ ]$ az vm create --resource-group CloudLabRG --name MyFirstVM --size Standard_B2als_v2 --image Ubuntu2404 --admin-username azuruser --generate-ssh-keys --public-ip-sku Standard --location italynorth
The default value of '--size' will be changed to 'Standard_D2s_v5' from 'Standard_DS1_v2' in a future release.
SSH key files '/home/dan/.ssh/id_rsa' and '/home/dan/.ssh/id_rsa.pub' have been generated under ~/.ssh to allow SSH access to the VM. If using machines without permanent storage, back up your keys to a safe location.
| Running ..
{
  "fqdns": "",
  "id": "/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/CloudLabRG/providers/Microsoft.Compute/virtualMachines/MyFirstVM",
  "location": "italynorth",
  "macAddress": "7C-ED-8D-01-51-5E",
  "powerState": "VM running",
  "privateIpAddress": "10.0.0.4",
  "publicIpAddress": "172.213.221.253",
  "resourceGroup": "CloudLabRG"
}
dan [ ~ ]$
dan [ ~ ]$
```

2.3.4 Verificare si Conectare

```
dan [ ~ ]$ az vm list-ip-addresses --resource-group CloudLabRG --name MyFirstVM --output table
VirtualMachine    PublicIPAddresses    PrivateIPAddresses
----------------  -------------------  --------------------
MyFirstVM         172.213.221.253      10.0.0.4
dan [ ~ ]$ ssh azuruser@172.213.221.253
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Dec 22 14:26:11 UTC 2025

  System load:  0.09              Processes:             122
  Usage of /:   5.6% of 28.02GB   Users logged in:       0
  Memory usage: 7%                IPv4 address for eth0: 10.0.0.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

2.3.5 Stergerea unui VM

```
dan [ ~ ]$ az vm delete --resource-group CloudLabRG --name MyFIrstVM --yes
dan [ ~ ]$
```

2.4 Scalabilitate si Load Balancing (VM Scale Sets)

2.4.1 Crearea unui cluster de VM-uri

```
dan [ ~ ]$ az vmss create --resource-group CloudLabRG --name LabScaleSet --image Ubuntu2404 --vm-sku Standard_B2als_v2 --upgrade-policy-mode automatic --admin-username azureuser --generate-ssh-keys --instance-count 2 --backend-port 80 --
-location italynorth
The default value of '--vm-sku' will be changed to 'Standard_D2s_v5' from 'Standard_DS1_v2' in a future release.
{
  "vmss": {
    "constrainedMaximumCapacity": false,
    "highSpeedInterconnectPlacement": "None",
    "orchestrationMode": "Flexible",
    "platformFaultDomainCount": 1,
    "provisioningState": "Succeeded",
    "singlePlacementGroup": false,
    "timeCreated": "2025-12-22T14:37:41.6442138+00:00",
    "uniqueId": "cb7582fc-018b-463e-883e-19215823bd4c",
    "upgradePolicy": {
      "mode": "Automatic"
    },
    "virtualMachineProfile": {
      "networkProfile": {
        "networkApiVersion": "2020-11-01",
        "networkInterfaceConfigurations": [
          {
            "name": "labsc67f3Nic",
            "properties": {
              "auxiliaryMode": "None",
              "auxiliarySku": "None",
              "deleteOption": "Delete",
              "disableIpgpStateTracking": false,
              "dnsSettings": {
                "dnsServers": []
              },
```

2.4.2 Demonstratia scalarii manual

```
dan [ ~ ]$ az vmss scale --resource-group CloudLabRG --name LabScaleSet --new-capacity 4
(OperationNotAllowed) Operation could not be completed as it results in exceeding approved Total Regional Cores quota. Additional details - Deployment Model: Resource Manager, Location: ItalyNorth, Current Limit: 6, Current Usage: 6, Additional Required: 2, (Minimum) New Limit Required: 8. Setup Alerts when Quota reaches threshold. Learn more at https://aka.ms/quotamonitoringalerting . Submit a request for Quota increase at https://aka.ms/ProdportalCRP/#blade/Microsoft_Azure_Capacity/UsageAndQuota.ReactView/Parameters/%7B%22subscriptionId%22:%22b8a400bd-6466-4af0-a961-5858bd830924%22,%22command%22:%22openQuotaApprovalBlade%22,%22quotas%22:[%7B%22location%22:%22ItalyNorth%22,%22providerId%22:%22Microsoft.Compute%22,%22resourceName%22:%22cores%22,%22quotaRequest%22:%7B%22properties%22:%7B%22limit%22:8,%22unit%22:%22Count%22,%22name%22:%7B%22value%22:%22cores%22%7D%7D%7D%7D]%7D by specifying parameters listed in the 'Details' section for deployment to succeed. Please read more about quota limits at https://docs.microsoft.com/en-us/azure/azure-supportability/regional-quota-requests
Code: OperationNotAllowed
Message: Operation could not be completed as it results in exceeding approved Total Regional Cores quota. Additional details - Deployment Model: Resource Manager, Location: ItalyNorth, Current Limit: 6, Current Usage: 6, Additional Required: 2, (Minimum) New Limit Required: 8. Setup Alerts when Quota reaches threshold. Learn more at https://aka.ms/quotamonitoringalerting . Submit a request for Quota increase at https://aka.ms/ProdportalCRP/#blade/Microsoft_Azure_Capacity/UsageAndQuota.ReactView/Parameters/%7B%22subscriptionId%22:%22b8a400bd-6466-4af0-a961-5858bd830924%22,%22command%22:%22openQuotaApprovalBlade%22,%22quotas%22:[%7B%22location%22:%22ItalyNorth%22,%22providerId%22:%22Microsoft.Compute%22,%22resourceName%22:%22cores%22,%22quotaRequest%22:%7B%22properties%22:%7B%22limit%22:8,%22unit%22:%22Count%22,%22name%22:%7B%22value%22:%22cores%22%7D%7D%7D%7D]%7D by specifying parameters listed in the 'Details' section for deployment to succeed. Please read more about quota limits at https://docs.microsoft.com/en-us/azure/azure-supportability/regional-quota-requests
Target: LabScaleSet_e5f91b37
dan [ ~ ]$ az vmss list-instances --resource-group CloudLabRG --name LabScaleSet --output table
InstanceId        Location    Name                ResourceGroup
----------------  ----------  ------------------  ---------------
LabScaleSet_4f928562  ItalyNorth  LabScaleSet_4f928562  CloudLabRG
LabScaleSet_86a4e406  ItalyNorth  LabScaleSet_86a4e406  CloudLabRG
LabScaleSet_8d3c5475  ItalyNorth  LabScaleSet_8d3c5475  CloudLabRG
LabScaleSet_e5f91b37  ItalyNorth  LabScaleSet_e5f91b37  CloudLabRG
dan [ ~ ]$
```

2.4.3 Stergerea resurselor

```
dan [ ~ ]$ az vmss delete --resource-group CloudLabRG --name LabScaleSet
dan [ ~ ]$ az vmss list-instances --resource-group CloudLabRG --name LabScaleSet --output table
(ParentResourceNotFound) Failed to perform 'read' on resource(s) of type 'virtualMachineScaleSets/virtualMachines', because the parent resource '/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/CloudLabRG/providers/Microsoft.Compute/virtualMachineScaleSets/LabScaleSet' could not be found.
Code: ParentResourceNotFound
Message: Failed to perform 'read' on resource(s) of type 'virtualMachineScaleSets/virtualMachines', because the parent resource '/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/CloudLabRG/providers/Microsoft.Compute/virtualMachineScaleSets/LabScaleSet' could not be found.
dan [ ~ ]$
```

2.5 Kubernetes (AKS)

2.5.1 Arhitectura unui cluster

2.5.2 Azure Kubernetes Service

2.5.3 Inregistrarea furnizorului de resurse

```
dan [ ~ ]$ az provider register --namespace Microsoft.ContainerService
Registering is still on-going. You can monitor using 'az provider show -n Microsoft.ContainerService'
dan [ ~ ]$ az provider show -n Microsoft.ContainerService --query "registrationState"
"Registering"
dan [ ~ ]$ az provider show -n Microsoft.ContainerService --query "registrationState"
"Registering"
dan [ ~ ]$ az provider show -n Microsoft.ContainerService --query "registrationState"
"Registered"
dan [ ~ ]$
```

2.5.4 Crearea unui cluster gestionat

```
dan [ ~ ]$ az aks create --resource-group CloudLabRG --name LabAKSCluster --node-count 1 --generate-ssh-keys --tier free
Default SSH key behavior will change. When no SSH key parameters are provided, the command will behave as if '--no-ssh-key' was passed instead of failing in next breaking change release(2.80.0) scheduled for Nov 2025.
{
  "aadProfile": null,
  "addonProfiles": null,
  "agentPoolProfiles": [
    {
      "availabilityZones": null,
      "capacityReservationGroupId": null,
      "count": 1,
      "creationData": null,
      "currentOrchestratorVersion": "1.33.5",
      "eTag": null,
      "enableAutoScaling": false,
      "enableEncryptionAtHost": false,
      "enableFips": false,
      "enableNodePublicIp": false,
      "enableUltraSsd": false,
      "gatewayProfile": null,
      "gpuInstanceProfile": null,
      "gpuProfile": null,
      "hostGroupId": null,
      "kubeletConfig": null,
      "kubeletDiskType": "OS",
      "linuxOsConfig": null,
      "maxCount": null,
      "maxPods": 250,
      "messageOfTheDay": null,
      "minCount": null,
      "mode": "System",
```

2.5.5 Conectarea terminalului la cluster

```
dan [ ~ ]$ az aks get-credentials --resource-group CloudLabRG --name LabAKSCluster
Merged "LabAKSCluster" as current context in /home/dan/.kube/config
dan [ ~ ]$
```

2.5.6 Lansarea unei Aplicatii Web (AKS Hello World)

```
dan [ ~ ]$ nano aks-app.yaml
dan [ ~ ]$ cat aks-app.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: aks-helloworld
spec:
  replicas: 1
  selector:
    matchLabels:
      app: aks-helloworld
  template:
    metadata:
      labels:
        app: aks-helloworld
    spec:
      nodeSelector:
        kubernetes.io/os: linux
      containers:
        - name: aks-helloworld
          # Imagine oficială Microsoft stabilă pentru testare
          image: mcr.microsoft.com/azuredocs/aks-helloworld:v1
          ports:
            - containerPort: 80
          env:
            - name: TITLE
              value: "Salutare din Laboratorul Azure!"
---
apiVersion: v1
kind: Service
metadata:
  name: aks-helloworld
spec:
  type: LoadBalancer
  ports:
```

```
dan [ ~ ]$ kubectl apply -f aks-app.yaml
deployment.apps/aks-helloworld created
service/aks-helloworld created
dan [ ~ ]$ kubectl get service azure-vote-front --watch
Error from server (NotFound): services "azure-vote-front" not found
dan [ ~ ]$ kubectl get service azure-vote-front --watch
Error from server (NotFound): services "azure-vote-front" not found
dan [ ~ ]$
```

2.5.7 Comenzi comune in Kubernetes

```
dan [ ~ ]$ kubectl get nodes
NAME                                STATUS   ROLES    AGE   VERSION
aks-nodepool1-71914792-vmss000000   Ready    <none>   11m   v1.33.5
dan [ ~ ]$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
aks-helloworld-658684fd99-t7lz9     1/1     Running   0          2m7s
dan [ ~ ]$ kubectl get pods-o wide
error: the server doesn't have a resource type "pods-o"
dan [ ~ ]$ kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP            NODE                                NOMINATED NODE   READINESS GATES
aks-helloworld-658684fd99-t7lz9     1/1     Running   0          2m36s   10.244.0.109  aks-nodepool1-71914792-vmss000000   <none>           <none>
dan [ ~ ]$ kubectl get services
NAME            TYPE           CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
aks-helloworld  LoadBalancer   10.0.157.247   <pending>     80:32055/TCP   2m57s
kubernetes      ClusterIP      10.0.0.1       <none>        443/TCP        14m
dan [ ~ ]$ kubectl describe pod aks-helloworld-658684fd99-t7lz9
Name:             aks-helloworld-658684fd99-t7lz9
Namespace:        default
Priority:         0
Service Account:  default
Node:             aks-nodepool1-71914792-vmss000000/10.224.0.4
Start Time:       Mon, 22 Dec 2025 15:18:10 +0000
Labels:           app=aks-helloworld
                  pod-template-hash=658684fd99
Annotations:      <none>
Status:           Running
IP:               10.244.0.109
IPs:
  IP:           10.244.0.109
Controlled By:  ReplicaSet/aks-helloworld-658684fd99
Containers:
  aks-helloworld:
    Container ID:   containerd://fc3dedc5366f347790e891c147797da6936ed3181131f2b8e4321d8cb3dc8012
```

```
dan [ ~ ]$ kubectl logs aks-helloworld-658684fd99-t7lz9
Checking for script in /app/prestart.sh
Running script /app/prestart.sh
Running inside /app/prestart.sh, you could add migrations to this file, e.g.:

#! /usr/bin/env bash

# Let the DB start
sleep 10;
# Run migrations
alembic upgrade head

/usr/lib/python2.7/dist-packages/supervisor/options.py:298: UserWarning: Supervisord is running as root and it is searching for its confi
specify a "-c" argument specifying an absolute path to a configuration file for improved security.
  'Supervisord is running as root and it is searching '
2025-12-22 15:18:18,820 CRIT Supervisor running as root (no user in config file)
2025-12-22 15:18:18,820 INFO Included extra file "/etc/supervisor/conf.d/supervisord.conf" during parsing
2025-12-22 15:18:18,827 INFO RPC interface 'supervisor' initialized
2025-12-22 15:18:18,827 CRIT Server 'unix_http_server' running without any HTTP authentication checking
2025-12-22 15:18:18,827 INFO supervisord started with pid 7
2025-12-22 15:18:19,829 INFO spawned: 'nginx' with pid 10
2025-12-22 15:18:19,831 INFO spawned: 'uwsgi' with pid 11
[uWSGI] getting INI configuration from /app/uwsgi.ini
[uWSGI] getting INI configuration from /etc/uwsgi/uwsgi.ini
*** Starting uWSGI 2.0.15 (64bit) on [Mon Dec 22 15:18:19 2025] ***
compiled with version: 6.3.0 20170516 on 04 February 2018 16:22:48
os: Linux-5.15.0-1101-azure #110-Ubuntu SMP Wed Nov 19 02:16:35 UTC 2025
nodename: aks-helloworld-658684fd99-t7lz9
machine: x86_64
clock source: unix
pcre jit disabled
detected number of CPU cores: 4
current working directory: /app
detected binary path: /usr/local/bin/uwsgi
```

2.5.8 Steregerea resurselor

```
dan [ ~ ]$ az group delete --name CloudLabRG --yes --no-wait
dan [ ~ ]$
```

3 Module optionale

3.1 Azure Functions (Serverless)

3.1.1 Exemplu practic: Serverless HTTP Endpoint

```
dan [ ~ ]$ RND=$RANDOM
GRP="ServerlessRG-$RND"
LOC="italynorth"
STO="funcstore$RND"
APP="MyLabFuncApp-$RND"
echo "Se creeaza resursele in grupul: $GRP..."
Se creeaza resursele in grupul: ServerlessRG-27569...
dan [ ~ ]$ az group create --name $GRP --location $LOC
{
  "id": "/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/ServerlessRG-27569",
  "location": "italynorth",
  "managedBy": null,
  "name": "ServerlessRG-27569",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
```

```
dan [ ~ ]$ az storage account create --name $STO --location $LOC --resource-group $GRP --sku Standard_LRS
{
  "accessTier": "Hot",
  "accountMigrationInProgress": null,
  "allowBlobPublicAccess": false,
  "allowCrossTenantReplication": null,
  "allowSharedKeyAccess": null,
  "allowedCopyScope": null,
  "azureFilesIdentityBasedAuthentication": null,
  "blobRestoreStatus": null,
  "creationTime": "2025-12-22T16:35:01.551928+00:00",
  "customDomain": null,
  "defaultToOAuthAuthentication": null,
  "dnsEndpointType": null,
  "dualStackEndpointPreference": null,
  "enableExtendedGroups": null,
  "enableHttpsTrafficOnly": true,
  "enableNfsV3": null,
  "encryption": {
    "encryptionIdentity": null,
    "keySource": "Microsoft.Storage",
    "keyVaultProperties": null,
    "requireInfrastructureEncryption": null,
    "services": {
      "blob": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2025-12-22T16:35:01.598802+00:00"
      },
      "file": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2025-12-22T16:35:01.598802+00:00"
      },
```

```
dan [ ~ ]$ az provider register --namespace Microsoft.Web
Registering is still on-going. You can monitor using 'az provider show -n Microsoft.Web'
dan [ ~ ]$ az provider show --namespace Microsoft.Web --query "registrationState"
"Registering"
dan [ ~ ]$ az provider show --namespace Microsoft.Web --query "registrationState"
"Registering"
dan [ ~ ]$ az provider show --namespace Microsoft.Web --query "registrationState"
"Registered"
dan [ ~ ]$
```

```
dan [ ~ ]$ az functionapp create --resource-group $GRP --consumption-plan-location $LOC --runtime node --runtime-version 24 --functions-version 4 --name $APP --storage-account $STO
Resource provider 'Microsoft.OperationalInsights' used by this operation is not registered. We are registering for you.


Registration succeeded.
Resource provider 'microsoft.insights' used by this operation is not registered. We are registering for you.
Registration succeeded.
Application Insights "MyLabFuncApp-29198" was created for this Function App. You can visit https://portal.azure.com/#resource/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/ServerlessRG-29198/providers/microsoft.insig
hts/components/MyLabFuncApp-29198/overview to view your Application Insights component
{
  "autoGeneratedDomainNameLabelScope": null,
  "availabilityState": "Normal",
  "clientAffinityEnabled": false,
  "clientAffinityPartitioningEnabled": null,
  "clientAffinityProxyEnabled": false,
  "clientCertEnabled": false,
  "clientCertExclusionPaths": null,
  "clientCertMode": "Required",
  "cloningInfo": null,
  "containerSize": 1536,
  "customDomainVerificationId": "73C9C20A8E9588CE61370D1678F676A3075253 2E403A33BBFAF6666E1B50FE2",
  "dailyMemoryTimeQuota": 0,
  "daprConfig": null,
  "defaultHostName": "mylabfuncapp-29198.azurewebsites.net",
  "dnsConfiguration": {
    "dnsAltServer": null,
    "dnsLegacySortOrder": null,
    "dnsMaxCacheTimeout": null,
    "dnsRetryAttemptCount": null,
    "dnsRetryAttemptTimeout": null,
    "dnsServers": null
  },
```

```
dan [ ~ ]$ FUNC_URL="https://$APP.azurewebsites.net"
echo "Aplicatia este activala: $FUNC_URL"
Aplicatia este activala: https://MyLabFuncApp-29198.azurewebsites.net
dan [ ~ ]$ curl-I $FUNC_URL
bash: curl-I: command not found
dan [ ~ ]$ curl -I $FUNC_URL
HTTP/2 200
content-type: text/html
date: Mon, 22 Dec 2025 16:57:43 GMT
content-length: 150093

dan [ ~ ]$
```

3.1.2 Adaugarea si publicarea codului personalizat

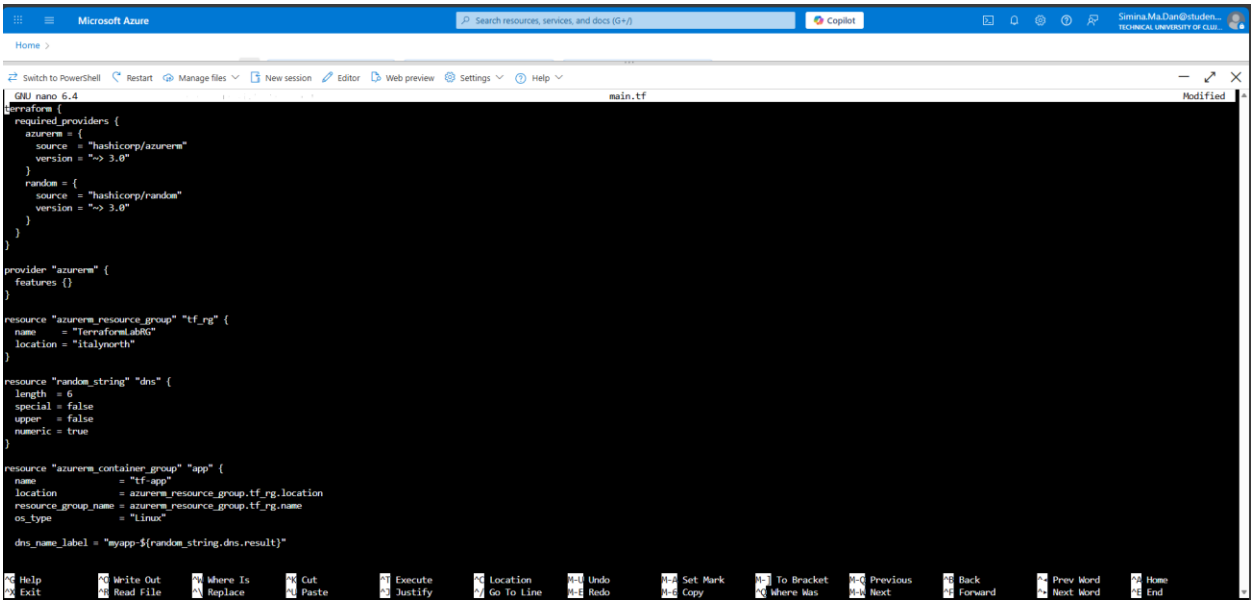
```
dan [ ~ ]$ mkdir MyLocalFunc
dan [ ~ ]$ cd MyLocalFunc
dan [ ~/MyLocalFunc ]$ func init --worker-runtime node --language javascript

The new Node.js programming model is generally available. Learn more at https://aka.ms/AzFuncNodeV4
Writing package.json
Writing .funcignore
Writing .gitignore
Writing host.json
Writing local.settings.json
Writing /home/dan/MyLocalFunc/.vscode/extensions.json
Running 'npm install'...dan [ ~/MyLocalFunc ]$
dan [ ~/MyLocalFunc ]$ func new --name MyCustomTrigger --template "HTTPtrigger" --authlevel anonymous
'local.settings.json' found in root directory (/home/dan/MyLocalFunc).
Resolving worker runtime to 'node'.
Select a number for template:HTTPtrigger
Function name: [httpTrigger] Creating a new file /home/dan/MyLocalFunc/src/functions/MyCustomTrigger.js
The function "MyCustomTrigger" was created successfully from the "HTTPtrigger" template.
dan [ ~/MyLocalFunc ]$ func azure functionapp publish $APP
'local.settings.json' found in root directory (/home/dan/MyLocalFunc).
Resolving worker runtime to 'node'.
Setting Functions site property 'netFrameworkVersion' to 'v8.0'
Getting site publishing info...
[2025-12-22T17:01:15.138Z] Starting the function app deployment...
Creating archive for current directory...
Uploading 657.63 KB [##############################################################################]
Upload completed successfully.
Deployment completed successfully.
[2025-12-22T17:01:43.407Z] Syncing triggers...
Functions in MyLabFuncApp-29198:
    MyCustomTrigger - [httpTrigger]
        Invoke url: https://mylabfuncapp-29198.azurewebsites.net/api/mycustomtrigger

dan [ ~/MyLocalFunc ]$ curl"https://$APP.azurewebsites.net/api/MyCustomTrigger?name=CloudLabUser"
bash: curlhttps://MyLabFuncApp-29198.azurewebsites.net/api/MyCustomTrigger?name=CloudLabUser: No such file or directory
dan [ ~/MyLocalFunc ]$ curl "https://$APP.azurewebsites.net/api/MyCustomTrigger?name=CloudLabUser"
dan [ ~/MyLocalFunc ]$ curl "https://$APP.azurewebsites.net/api/MyCustomTrigger?name=CloudLabUser"
Hello, CloudLabUser!dan [ ~/MyLocalFunc ]$
```

3.1.3 Stergerea resurselor

```
dan [ ~/MyLocalFunc ]$ az group delete --name $GRP --yes --no-wait
dan [ ~/MyLocalFunc ]$
```

3.2 Terraform(IaC)

3.2.1 Task: Deploy Container ”HelloWorld”



3.2.2 Aplicare

```
dan [ ~/terraform ]$ nano main.tf
dan [ ~/terraform ]$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/random versions matching "~> 3.0"...
- Finding hashicorp/azurerm versions matching "~> 3.0"...
- Installing hashicorp/random v3.7.2...
- Installed hashicorp/random v3.7.2 (signed by HashiCorp)
- Installing hashicorp/azurerm v3.117.1...
- Installed hashicorp/azurerm v3.117.1 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
dan [ ~/terraform ]$ terraform apply-auto-approve
Terraform has no command named "apply-auto-approve".

To see all of Terraform's top-level commands, run:
  terraform -help

dan [ ~/terraform ]$ terraform apply -auto-approve


Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # azurerm_container_group.app will be created
  + resource "azurerm_container_group" "app" {
      + dns_name_label               = (known after apply)
```

```
Outputs:

url = "http://myapp-ubw87f.italynorth.azurecontainer.io"
dan [ ~/terraform ]$
dan [ ~/terraform ]$ curl http://myapp-ubw87f.italynorth.azurecontainer.io/
<html>
<head>
  <title>Welcome to Azure Container Instances!</title>
</head>
<style>
h1 {
    color: darkblue;
    font-family:arial, sans-serif;
    font-weight: lighter;
}
</style>

<body>

<div align="center">
<h1>Welcome to Azure Container Instances!</h1>

<svg id="Layer_1" data-name="Layer 1" xmlns="http://www.w3.org/2000/svg" viewBox="0 0 49.8 49.9" width="250px" height="250px">
  <title>ContainerInstances_rgb_UI</title>
  <path d="M41.9,11.368A11.929,11.929,0,0,0,20.3,5.061a9.444,9.444,0,0,0-14.932,9.8A8.969,8.969,0,0,0,9.064,32H39.442A10.463,10.463,0,0,0,41.9,11.368Z" transform="translate(-0.1 -0.1)" fill="#fff"/>
        <path d="M41.9,11.368A11.929,11.929,0,0,0,20.3,5.061a9.444,9.444,0,0,0-14.932,9.8A8.969,8.969,0,0,0,9.064,32H39.442A10.463,10.463,0,0,0,41.9,11.368Z" transform="translate(-0.1 -0.1)" fill="#27a9e1" opacity="0.6" style="isolation
:isolate"/>
        <path d="M13,22a1,1,0,0,0-1,1V49a1,1,0,0,0,1,1H37a1,1,0,0,0,1-1V23a1,1,0,0,0-1-1Z" transform="translate(-0.1 -0.1)" fill="#672a7a"/>
        <path d="M26.95,16" transform="translate(-0.1 -0.1)" fill="none"/>
        <path d="M34.95,20" transform="translate(-0.1 -0.1)" fill="none"/>
        <polygon points="22.9 21.9 22.9 14.9 19.9 14.9 24.9 7.9 29.9 14.9 26.9 14.9 26.9 21.9 22.9 21.9" fill="#fff"/>
        <path d="M26.95,16" transform="translate(-0.1 -0.1)" fill="#814a98"/>
        <path d="M33,25H15V47H35V25ZM21,45H17V27h4Zm6,0H23V27h4Zm6,0H29V27h4Z" transform="translate(-0.1 -0.1)" fill="#b92025" opacity="0.3" style="isolation:isolate"/>
        <path d="M33,25H15V47H35V25ZM21,45H17V27h4Zm6,0H23V27h4Zm6,0H29V27h4Z" transform="translate(-0.1 -0.1)" fill="#fff" style="isolation:isolate"/>
</svg>
</div>

</body>
</html>
dan [ ~/terraform ]$
```

3.2.3 Stergerea resurselor

```
dan [ ~/terraform ]$ terraform destroy -auto-approve
random_string.dns: Refreshing state... [id=ubw87f]
azurerm_resource_group.tf_rg: Refreshing state... [id=/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/TerraformLabRG]
azurerm_container_group.app: Refreshing state... [id=/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/TerraformLabRG/providers/Microsoft.ContainerInstance/containerGroups/tf-app]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # azurerm_container_group.app will be destroyed
  - resource "azurerm_container_group" "app" {
      - dns_name_label             = "myapp-ubw87f" -> null
      - dns_name_label_reuse_policy = "Unsecure" -> null
      - exposed_port               = [
          - {
              - port     = 80
              - protocol = "TCP"
            },
        ] -> null
      - fqdn                       = "myapp-ubw87f.italynorth.azurecontainer.io" -> null
      - id                         = "/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/TerraformLabRG/providers/Microsoft.ContainerInstance/containerGroups/tf-app" -> null
      - ip_address                 = "4.232.4.8" -> null
      - ip_address_type            = "Public" -> null
      - location                   = "italynorth" -> null
      - name                       = "tf-app" -> null
      - os_type                    = "Linux" -> null
      - resource_group_name        = "TerraformLabRG" -> null
      - restart_policy             = "Always" -> null
      - sku                        = "Standard" -> null
      - subnet_ids                 = [] -> null
      - tags                       = {} -> null
      - zones                      = [] -> null
        # (1 unchanged attribute hidden)

      - container {
          - commands                     = [] -> null
          - cpu                          = 0.5 -> null
          - cpu_limit                    = 0 -> null
          - environment_variables        = {} -> null
```

3.3 Ansible (Configuration Management)

3.3.1 Provizionarea infrastructurii

```
dan [ ~/ansible ]$ az group create --name CloudLabRG --location italynorth
{
  "id": "/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/CloudLabRG",
  "location": "italynorth",
  "managedBy": null,
  "name": "CloudLabRG",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
dan [ ~/ansible ]$ az vm create --resource-group CloudLabRG --name AnsibleTargetVM --image Ubuntu2404 --size Standard_B2als_v2 --location italynorth --admin-username azureuser --generate-ssh-keys --public
a Standard
The default value of '--size' will be changed to 'Standard_D2s_v5' from 'Standard_DS1_v2' in a future release.
{
  "fqdns": "",
  "id": "/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/CloudLabRG/providers/Microsoft.Compute/virtualMachines/AnsibleTargetVM",
  "location": "italynorth",
  "macAddress": "7C-ED-8D-01-7A-F1",
  "powerState": "VM running",
  "privateIpAddress": "10.0.0.4",
  "publicIpAddress": "172.213.247.28",
  "resourceGroup": "CloudLabRG"
}
dan [ ~/ansible ]$ az vm open-port --resource-group CloudLabRG --name AnsibleTargetVM --port 80 --priority 1010
{
  "defaultSecurityRules": [
    {
      "access": "Allow",
      "description": "Allow inbound traffic from all VMs in VNET",
      "destinationAddressPrefix": "VirtualNetwork",
      "destinationAddressPrefixes": [],
      "destinationPortRange": "*",
      "destinationPortRanges": [],
      "direction": "Inbound",
      "etag": "W/\"f6cdf0e0-4069-4170-a196-2a1cf6f2fd45\"",
      "id": "/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/CloudLabRG/providers/Microsoft.Network/networkSecurityGroups/AnsibleTargetVMNSG/defaultSecurityRules/AllowVnetInBound",
      "name": "AllowVnetInBound",
      "priority": 65000,
      "protocol": "*",
      "provisioningState": "Succeeded",
      "resourceGroup": "CloudLabRG",
      "sourceAddressPrefix": "VirtualNetwork",
      "sourceAddressPrefixes": [],
```

3.3.2 Definirea inventarului

```
dan [ ~/ansible ]$ nano inventory.ini
dan [ ~/ansible ]$ cat inventory.ini
[webservers]
172.213.247.28 ansible_user=azureuser ansible_ssh_private_key_file=~/.ssh/id_rsa
dan [ ~/ansible ]$
```

3.3.3 Automatizare – Web Server & System Hardening

```
dan [ ~/ansible ]$ nano setup_server.yml
dan [ ~/ansible ]$ cat setup_server.yml
- name: Hardening & Web Server Setup
  hosts: webservers
  become: true  # Rulează comenzile cu sudo

  vars:
    app_user: "webapp_user"
    http_port: 80

  tasks:
    # --- 1. System Hardening ---
    - name: Create application user
      user:
        name: "{{ app_user }}"
        state: present
        shell: /bin/bash

    - name: Allow SSH traffic (CRITICAL before enabling UFW)
      ufw:
        rule: allow
        name: OpenSSH

    - name: Allow HTTP traffic
      ufw:
        rule: allow
        port: "{{ http_port }}"
        proto: tcp

    - name: Enable firewall (UFW)
      ufw:
        state: enabled
        policy: deny

    # --- 2. Application Configuration ---
    - name: Install Nginx
      apt:
        name: nginx
        state: present
        update_cache: yes

    - name: Start Nginx service
      service:
        name: nginx
        state: started
        enabled: yes
```

3.3.4 Executie si validare

```
dan [ ~/ansible ]$ ANSIBLE_HOST_KEY_CHECKING=False ansible-playbook -i inventory.ini setup_server.yml

PLAY [Hardening & Web Server Setup] ***********************************************************************************************************

TASK [Gathering Facts] ************************************************************************************************************************
[WARNING]: Host '172.213.247.28' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation of another Python interpreter could cause a different interpreter
See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [172.213.247.28]

TASK [Create application user] ****************************************************************************************************************
changed: [172.213.247.28]

TASK [Allow SSH traffic (CRITICAL before enabling UFW)] ***************************************************************************************
changed: [172.213.247.28]

TASK [Allow HTTP traffic] *********************************************************************************************************************
changed: [172.213.247.28]

TASK [Enable firewall (UFW)] ******************************************************************************************************************
changed: [172.213.247.28]

TASK [Install Nginx] **************************************************************************************************************************
changed: [172.213.247.28]

TASK [Start Nginx service] ********************************************************************************************************************
ok: [172.213.247.28]

PLAY RECAP ************************************************************************************************************************************
172.213.247.28             : ok=7    changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

dan [ ~/ansible ]$ curl 172.213.247.28
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>
```

3.3.5 Stergerea resurselor

```
dan [ ~/ansible ]$ az group delete --name CloudLabRG --yes --no-wait
dan [ ~/ansible ]$
```

3.4 Jenkins (CI/CD Clasic)

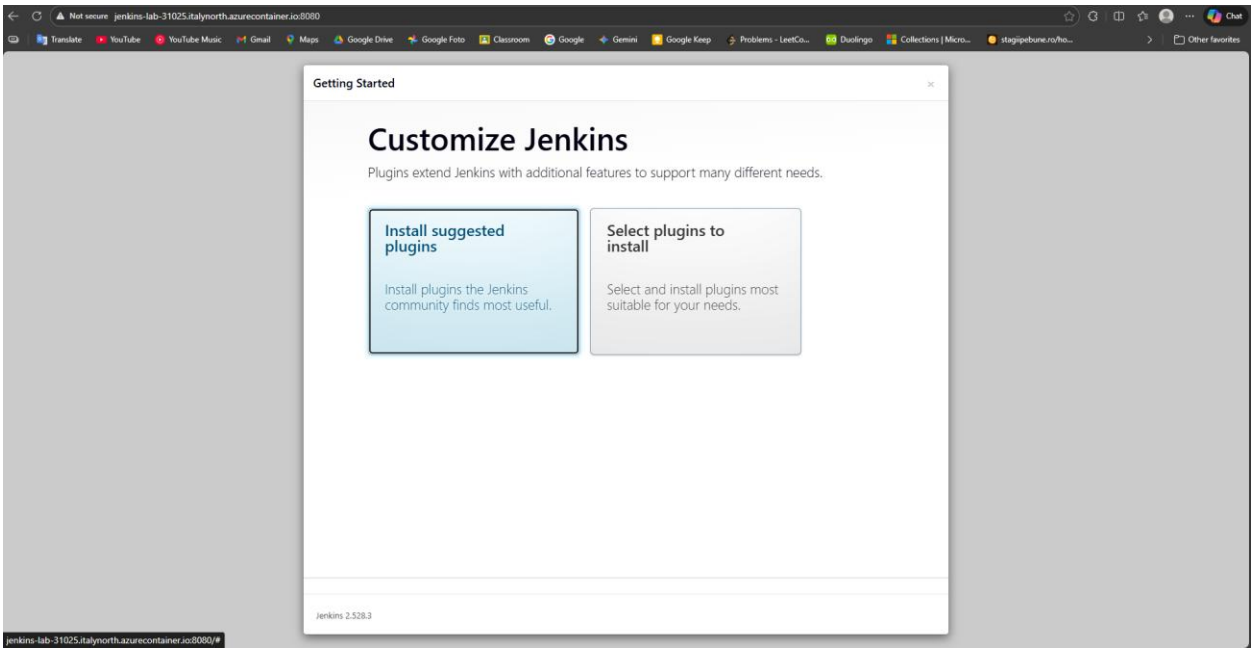
3.4.1 Jenkins in ecosistemul Azure

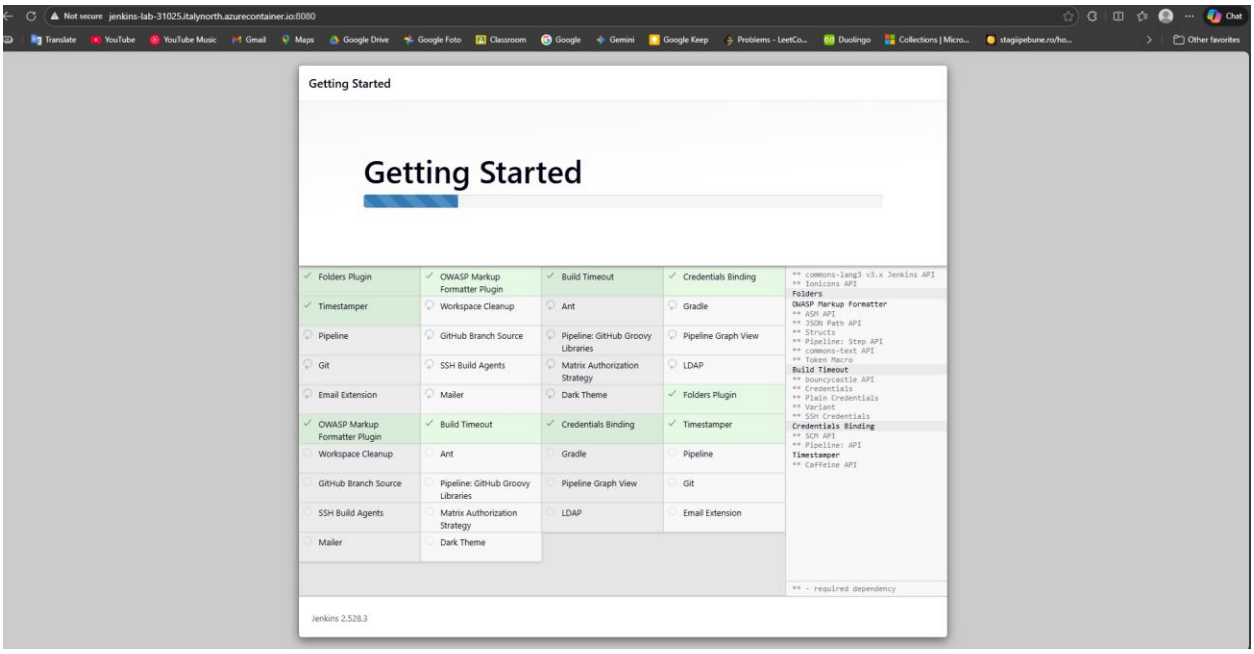
3.4.2 Lansarea Jenkins in Cloud (Azure Container Instances)

```
dan [ ~/jenkins ]$ az group create --name CloudLabRG --location italynorth
{
  "id": "/subscriptions/b8a400bd-6466-4af0-a961-5858bd830924/resourceGroups/CloudLabRG",
  "location": "italynorth",
  "managedBy": null,
  "name": "CloudLabRG",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
dan [ ~/jenkins ]$ az container create --resource-group CloudLabRG --name lab-jenkins --image jenkins/jenkins:lts --ports 8080 --dns-name-label jenkins-lab-${RANDOM} --cpu 1 --memory 1.5 --os-type Linux
{
  "confidentialComputeProperties": null,
  "containerGroupProfile": null,
  "containers": [
    {
      "command": null,
      "configMap": {
        "keyValuePairs": {}
      },
      "environmentVariables": [],
      "image": "jenkins/jenkins:lts",
      "instanceView": {
        "currentState": {
          "detailStatus": "",
          "exitCode": null,
          "finishTime": null,
          "startTime": "2025-12-22T18:01:45.599000+00:00",
          "state": "Running"
        },
        "events": [
          {
            "count": 1,
            "firstTimestamp": "2025-12-22T18:01:16+00:00",
            "lastTimestamp": "2025-12-22T18:01:16+00:00",
            "message": "pulling image \"jenkins/jenkins@sha256:b768bbb699b5f1b6579e23e7253ae1157a339a20a2e9befdf8e03923e893b800\"",
            "name": "Pulling",
            "type": "Normal"
          },
          {
            "count": 1
```

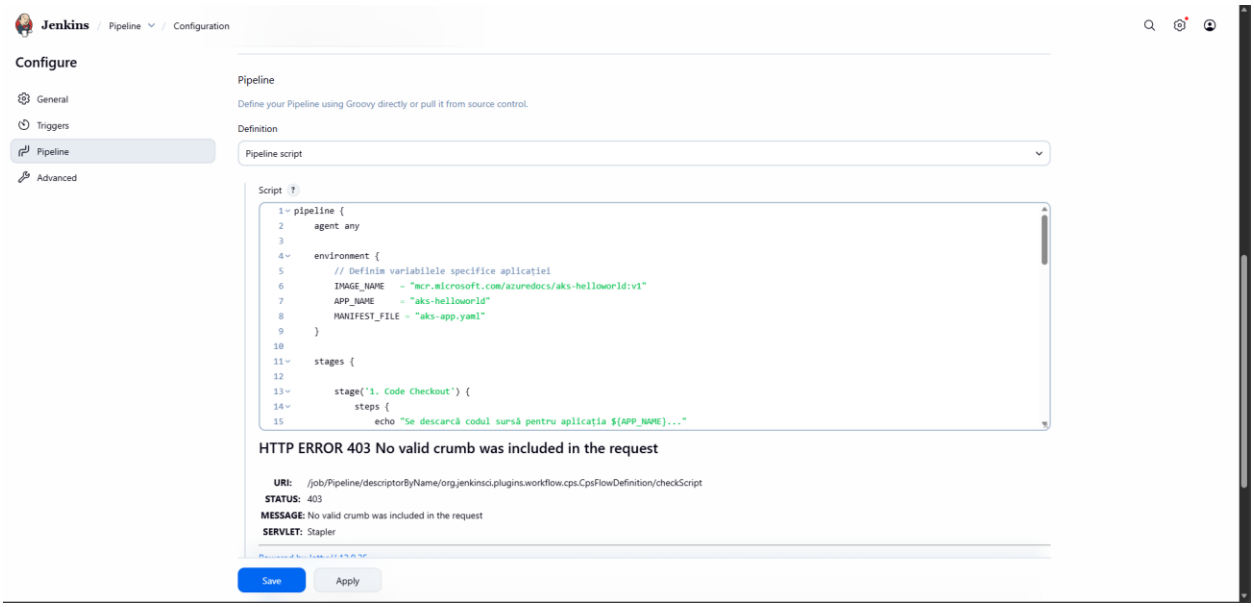
3.4.3 Accesare si configurare initiala

```
dan [ ~/jenkins ]$ az container show --resource-group CloudLabRG --name lab-jenkins --query ipAddress.fqdn --output tsv
jenkins-lab-31025.italynorth.azurecontainer.io
dan [ ~/jenkins ]$ az container logs --resource-group CloudLabRG --name lab-jenkins
[WARN  tini (19)] Tini is not running as PID 1 and isn't registered as a child subreaper.
Zombie processes will not be re-parented to Tini, so zombie reaping won't work.
To fix the problem, use the -s option or set the environment variable TINI_SUBREAPER to register Tini as a child subreaper, or run Tini as PID 1.
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2025-12-22 18:01:47.262+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2025-12-22 18:01:49.083+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2025-12-22 18:01:49.203+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e2420576
2025-12-22 18:01:50.358+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.J
2025-12-22 18:01:50.509+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2025-12-22 18:01:51.523+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.ge
2025-12-22 18:01:51.801+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@3c782d8e{Jenkins v2.528.3
=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@572e6fd9{STARTED}}
2025-12-22 18:01:51.848+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7bca6fac{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2025-12-22 18:01:51.889+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@3668d4{STARTING}[12.0.25,sto=0] @6201ms
2025-12-22 18:01:51.892+0000 [id=25]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2025-12-22 18:01:52.152+0000 [id=24]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.528.3
2025-12-22 18:01:52.407+0000 [id=31]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2025-12-22 18:01:52.448+0000 [id=31]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2025-12-22 18:01:54.635+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2025-12-22 18:01:54.641+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2025-12-22 18:01:54.642+0000 [id=31]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2025-12-22 18:01:55.125+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2025-12-22 18:01:55.126+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2025-12-22 18:01:55.126+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2025-12-22 18:01:55.130+0000 [id=32]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2025-12-22 18:01:56.113+0000 [id=46]    INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2025-12-22 18:01:56.622+0000 [id=31]    INFO    jenkins.install.SetupWizard#init:
[LF]>
[LF]> *************************************************************
[LF]> *************************************************************
[LF]> *************************************************************
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> 7504e53e90ea49ebbcf0b17b1a49b51f
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *************************************************************
[LF]> *************************************************************
```

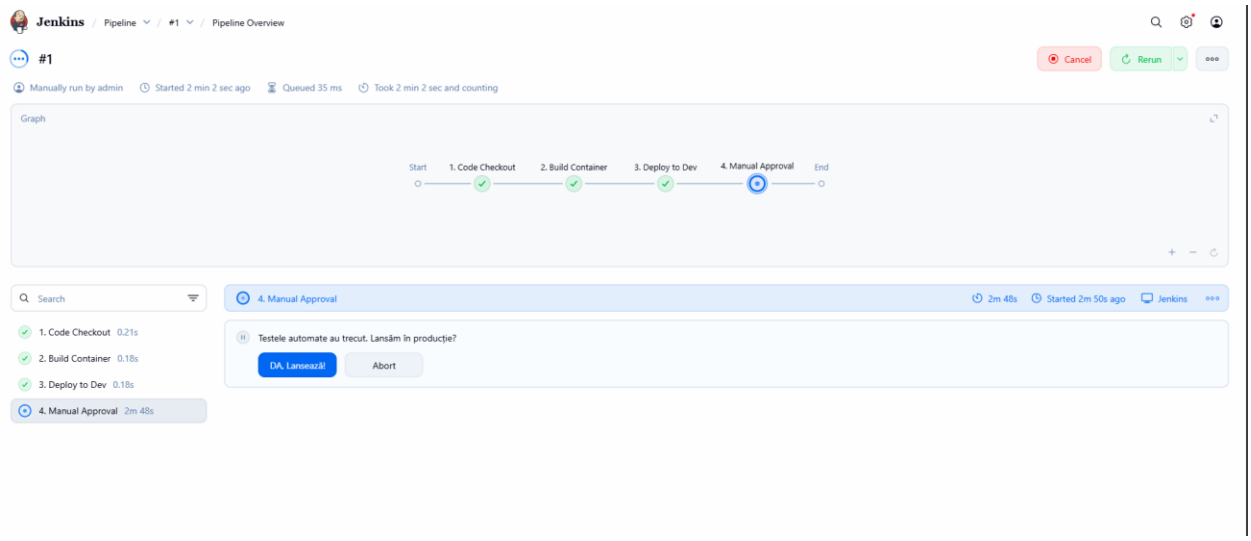


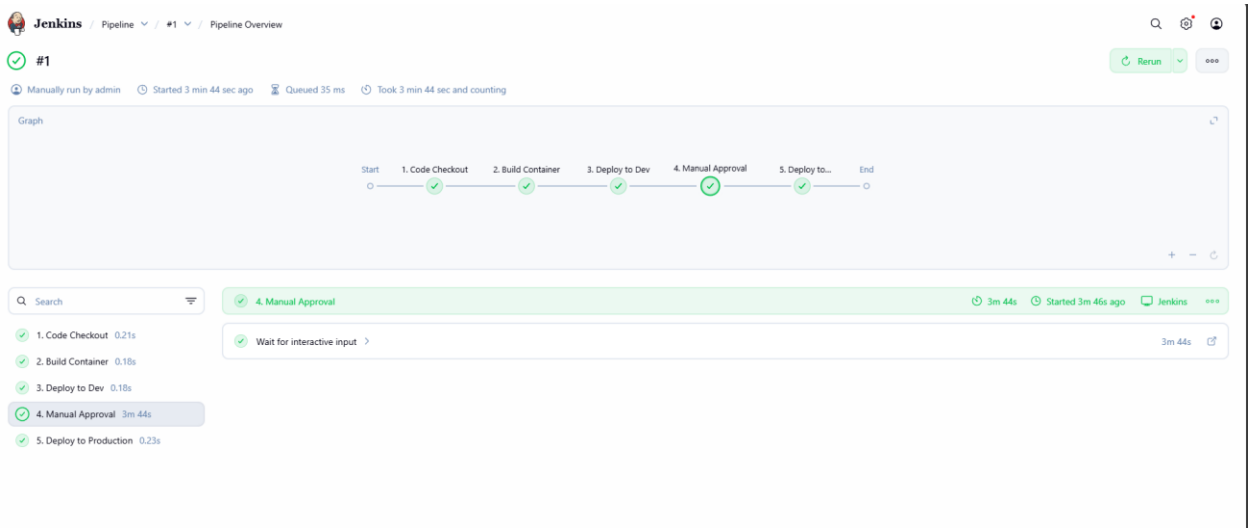


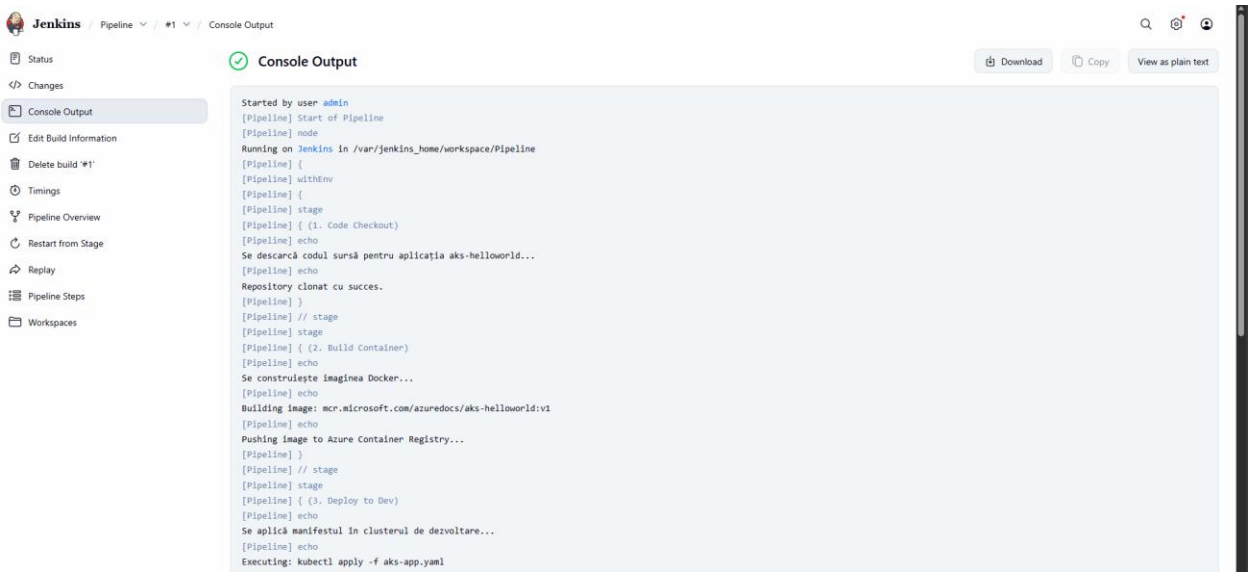
3.4.4 Simulare Pipeline Azure(Groovy)



3.4.5 Executia si monitorizarea pipeline-ului







3.4.6 Stergerea resurselor

```
dan [ ~ ]$ az container delete --resource-group CloudLabRG --name lab-jenkins --yes
{
  "confidentialComputeProperties": null,
  "containerGroupProfile": null,
  "containers": [
    {
      "command": null,
      "configMap": {
        "keyValuePairs": {}
      },
      "environmentVariables": [],
      "image": "jenkins/jenkins:lts",
      "instanceView": {
        "currentState": {
          "detailStatus": "",
          "exitCode": null,
          "finishTime": null,
          "startTime": "2025-12-22T18:01:45.599000+00:00",
          "state": "Running"
        },
        "events": [
          {
            "count": 1,
            "firstTimestamp": "2025-12-22T18:01:16+00:00",
            "lastTimestamp": "2025-12-22T18:01:16+00:00",
            "message": "pulling image \"jenkins/jenkins@sha256:b768bbb699b5f1b6579e23e7253ae1157a339a20a2e9befdf8e03923e893b800\"",
            "name": "Pulling",
            "type": "Normal"
          },
          {
            "count": 1,
            "firstTimestamp": "2025-12-22T18:01:36+00:00",
            "lastTimestamp": "2025-12-22T18:01:36+00:00",
```

```
dan [ ~ ]$ az group delete --name CloudLabRG --yes --no-wait
dan [ ~ ]$
```

3.5 GitHub Actions

3.6 Stiva ELK & Grafana

3.6.1 Configurare

Welcome | logstash.conf | docker-compose.yml

logstash.conf

```
input {
  http {
    port => 5000
    codec => json
  }
}

filter {
  # Dacă nu există un timestamp, adăugăm unul bazat pe timpul curent
  if ![timestamp] {
    mutate {
      add_field => {
        "timestamp" => "%{@timestamp}"
      }
    }
  }
}

output {
  elasticsearch {
    hosts => ["elasticsearch:9200"]
    index => "ecommerce-logs-%{+YYYY.MM.dd}"
  }

  # Afișează logurile și în consolă pentru debugging
  stdout {
    codec => rubydebug
  }
}
```

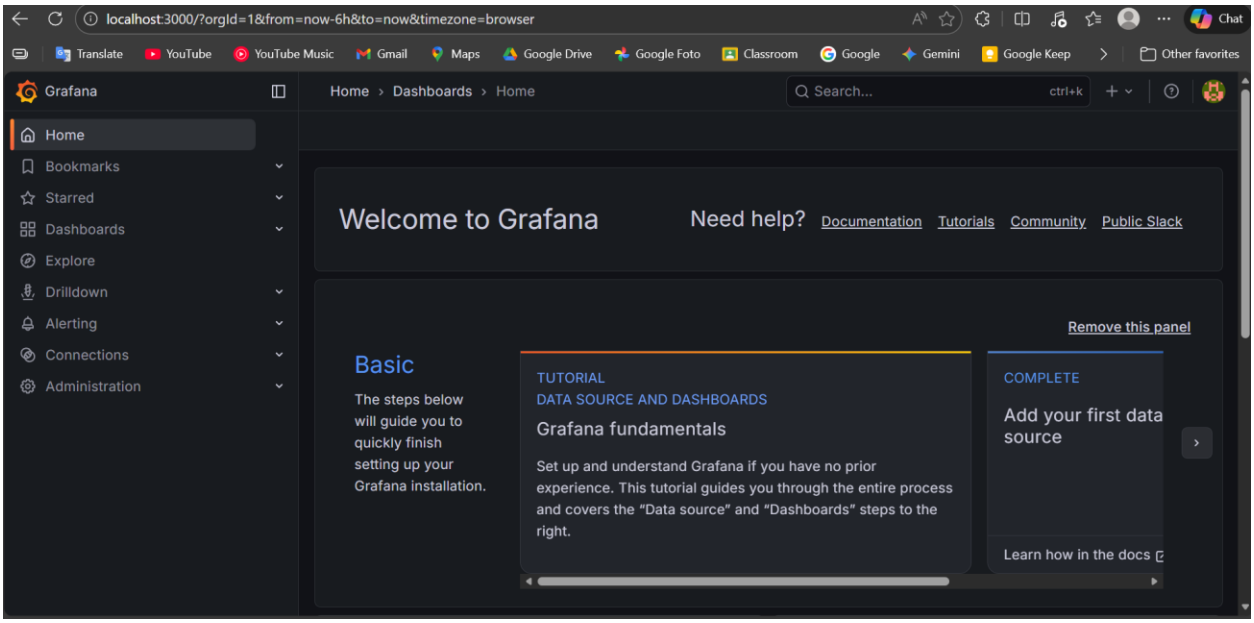
Welcome | logstash.conf | docker-compose.yml

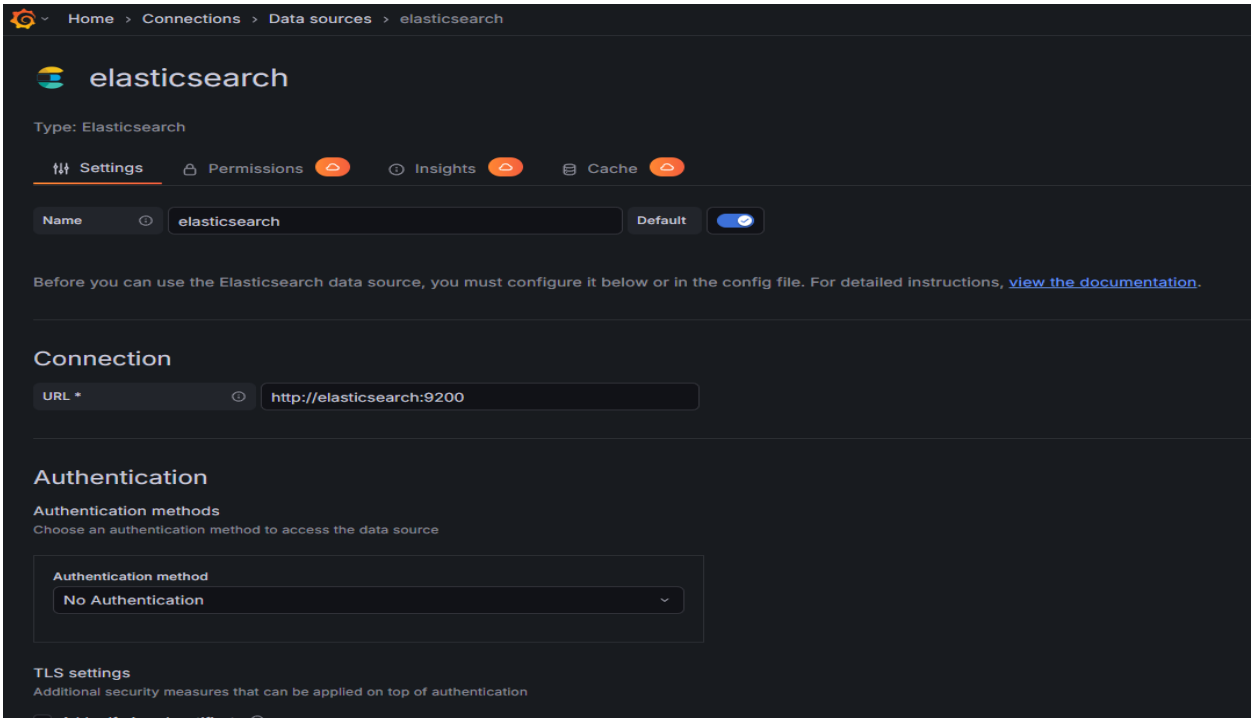
docker-compose.yml

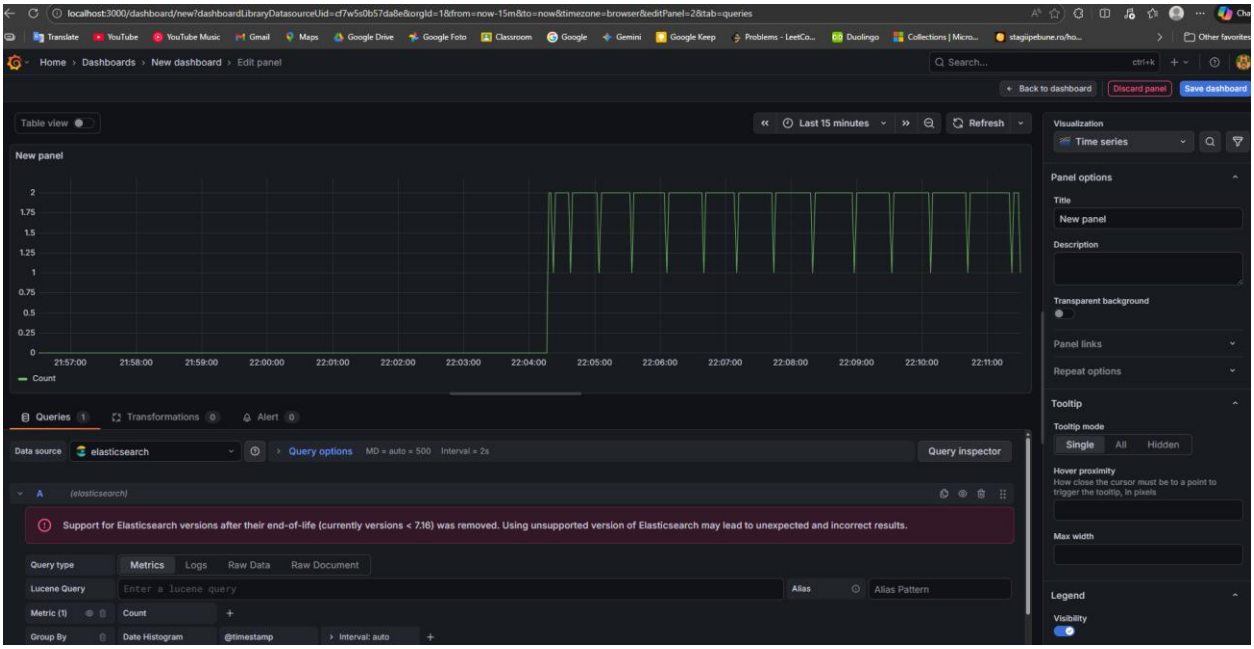
```yaml
version: "3.8"

services:
  elasticsearch:
    image: docker.elastic.co/elasticsearch/elasticsearch:7.9.2
    environment:
      - discovery.type=single-node
    ports:
      - "9200:9200"

  logstash:
    image: docker.elastic.co/logstash/logstash:7.9.2
    volumes:
      - ./logstash.conf:/usr/share/logstash/pipeline/logstash.conf
    ports:
      - "5000:5000"
    depends_on:
      - elasticsearch

  grafana:
    image: grafana/grafana:latest
    ports:
      - "3000:3000"
    depends_on:
      - elasticsearch

  traffic-generator:
    image: alpine
    depends_on:
      - logstash
    command: >
      sh -c '
      apk add --no-cache curl &&
      while true; do
        AMOUNT=$$((RANDOM % 500));
        USER_ID=$$((RANDOM % 100));
        curl -X POST -H "Content-Type: application/json" \
          -d "{\"user\":\"user_$$USER_ID\",\"amount\": $$AMOUNT}" \
          http://logstash:5000;
        sleep 1;
      done
      '
```

3.6.2 Rulare si vizualizare

```
PS D:\VirtualBox Shared\aso_share\Operating-Systems-Administration\Laboratory13\lab> docker-compose up -d
time="2025-12-22T22:03:42+02:00" level=warning msg="D:\\VirtualBox Shared\\aso_share\\Operating-Systems-Administration\\Laboratory1
move it to avoid potential confusion"
[+] Running 4/4
 ✓ Container lab-elasticsearch-1      Started                                                    0.5s
 ✓ Container lab-grafana-1            Started                                                    0.5s
 ✓ Container lab-logstash-1           Started                                                    0.5s
 ✓ Container lab-traffic-generator-1  Started                                                    0.4s
PS D:\VirtualBox Shared\aso_share\Operating-Systems-Administration\Laboratory13\lab>
```







3.6.3 Oprire resurse

```
PS D:\VirtualBox Shared\aso_share\Operating-Systems-Administration\Laboratory13\lab> docker-compose down
time="2025-12-22T22:14:08+02:00" level=warning msg="D:\\VirtualBox Shared\\aso_share\\Operating-Systems-Administration\\Laboratory13\\lab\\doc
move it to avoid potential confusion"
[+] Running 5/5
 ✓ Container lab-traffic-generator-1  Removed
 ✓ Container lab-grafana-1            Removed
 ✓ Container lab-logstash-1           Removed
 ✓ Container lab-elasticsearch-1      Removed
 ✓ Network lab_default                Removed
PS D:\VirtualBox Shared\aso_share\Operating-Systems-Administration\Laboratory13\lab>
```